Reflective Report

Group 9

This project helped us understand the course better with practical experience. We started solving real-life cases ourselves, though AI helped us dive deeper into concepts and finalize our work. We started by identifying key entities such as projects, employees, tasks, and clients, and then defined their relationships. Creating the database schema from scratch helped us understand how to properly structure data.

One of the most important aspects of the project was learning how to design tables, attributes, and relationships independently. Instead of relying on predefined schemas, we built the entire structure ourselves, including primary keys, foreign keys, and constraints. This significantly improved our understanding of how relational databases are constructed and how different tables interact with each other.

The main tool for designing the ERD was draw.io which triggered us a lot at the beginning but through experience and patience we got used to it. This helped us visualize the relationships between entities and made it easier to identify design mistakes early. Working with diagrams improved our ability to think conceptually before moving to implementation.

From a technical perspective, we strengthened our SQL skills using PostgreSQL. We wrote DDL scripts to create tables and enforce constraints, and we used SQL features such as TRIM to clean and standardize textual data, COALESCE to handle missing (NULL) values, and INTO in PL/pgSQL to store query results in variables, and NEW in trigger functions to reference the newly inserted row's column values, which were new concepts for us and expanded our understanding of data processing and procedural logic.

Another key learning outcome was working with large datasets. We used Python and the Faker library to generate realistic data for testing purposes which turned out to be tougher than we expected. We tried a lot to get a dataset which contained logical values and also we needed those values to work between the tables correspondingly which was hard (we spent 7 nights and 7 days). This allowed us to populate the database with thousands of records and better simulate real-world conditions. It also helped us understand how data generation tools can support database development and testing. And we understood that having real data is a huge advantage.

We also implemented more advanced database features such as triggers, stored procedures, and indexes. These allowed us to enforce business rules directly within the database and improve performance for frequently used queries. For example, triggers were used to prevent invalid operations, while indexes helped optimize query execution time.

Throughout the project, we faced several challenges. One of the main difficulties was maintaining consistency across different components, such as the ERD, schema, and query logic. Even small mismatches in table or attribute names could cause errors. Another significant challenge was generating and inserting large amounts of realistic data for DML script. Ensuring that all foreign key relationships were satisfied and that the data remained logically consistent across tables required careful planning and debugging. Additionally, it was a challenge to design realistic business rules and ensure that all parts of the system aligned with them.

In conclusion, this project provided valuable hands-on experience in database design and implementation. We improved both our technical and analytical skills, including SQL programming, data modeling, and performance optimization. Overall, the project helped us better understand how databases support real-world systems and prepared us for more advanced work in this field.